

The Singular Value Decomposition

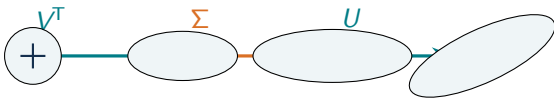
A geometric interpretation in two dimensions

Every linear map is a rotation, a coordinate-wise scaling, and another rotation.

The factorization

$$A = U\Sigma V^T$$

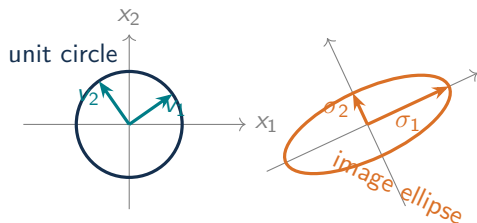
- ▶ V^T changes into the right-singular-vector coordinates.
- ▶ Σ stretches the coordinate axes by σ_1, σ_2 .
- ▶ U rotates into the left-singular-vector coordinates.



The unit circle becomes an ellipse

For any unit vector x , the point Ax lies on an ellipse. The longest and shortest radii of that ellipse are the singular values.

$$\sigma_1 = \max_{\|x\|=1} \|Ax\|, \quad \sigma_2 = \min_{\|x\|=1} \|Ax\|$$

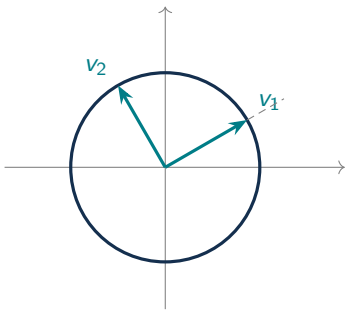


Right singular vectors choose input directions

The columns of $V = [v_1 \ v_2]$ form an orthonormal basis of the input space.

$$Av_i = \sigma_i u_i$$

Each v_i identifies an input direction that A maps exactly onto a principal axis of the output ellipse.



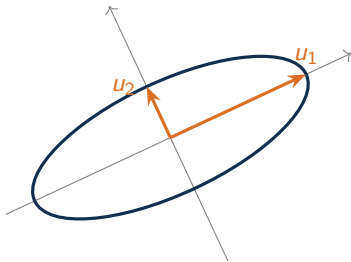
orthogonal input directions

Left singular vectors choose output directions

The columns of $U = [u_1 \ u_2]$ point along the orthonormal principal axes of the output ellipse.

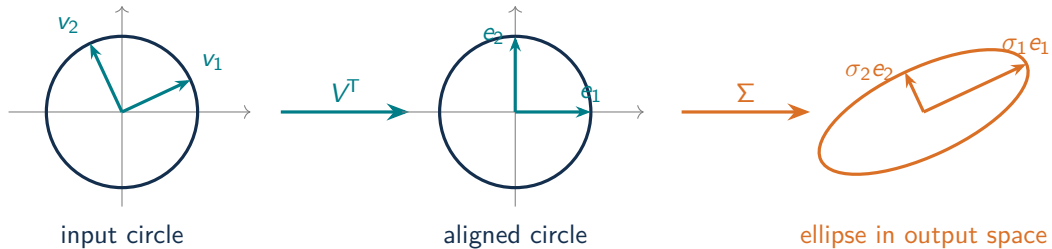
$$Av_1 = \sigma_1 u_1, \quad Av_2 = \sigma_2 u_2$$

The direction can rotate, while the length changes by the corresponding singular value.



orthogonal output directions

A complete geometric picture

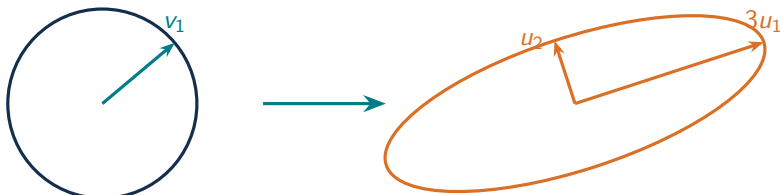


A numerical example

Let

$$A = U \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} V^T.$$

Then A maps the unit circle to an ellipse with semi-axis lengths 3 and 1.



The largest amplification occurs along v_1 ; the weakest occurs along v_2 .

Eigenvalues and singular values

	Eigenvalues	Singular values
Matrix	Defined only for square matrices	Defined for every $m \times n$ matrix
Equation	$Ax = \lambda x$	$Av_i = \sigma_i u_i$
Values	May be real or complex	Always real and nonnegative
Directions	Real eigenvectors may fail to exist. A matrix may also lack a full eigenvector basis.	Left and right singular-vector bases always exist
Meaning	A direction that stays on its own line under A	Orthogonal input directions with independent stretch factors

Important distinction

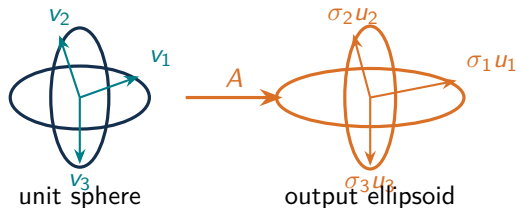
For a full-rank matrix, every singular value is positive. A rank-deficient matrix has one or more zero singular values.

The same geometry in three dimensions

In three dimensions, the unit sphere maps to an ellipsoid.

$$A = U \text{diag}(\sigma_1, \sigma_2, \sigma_3) V^T$$

The right singular vectors v_1, v_2, v_3 select orthogonal directions on the input sphere. The left singular vectors u_1, u_2, u_3 give the orthogonal axes of the output ellipsoid.



Why the SVD matters

- ▶ **Geometry:** It identifies the directions that a linear map stretches most and least.
- ▶ **Stability:** A very small σ_i marks a direction that the map nearly collapses.
- ▶ **Approximation:** Keeping the largest singular values gives the best low-rank approximation in Euclidean norm.

One sentence

The SVD expresses a matrix as orthogonal input directions, independent stretches, and orthogonal output directions.